

SAN LUIS, Argentina

WMO: 874360

Lat: 33.2744S

Lon: 66.3514W

Elev: 2339

StdP: 13.50

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.7	35.7	9.6	9.8	51.8	14.1	12.3	51.6	32.8	56.4	29.0	56.0	5.8	50	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.8	95.0	69.1	92.4	68.2	90.0	67.5	73.5	87.9	72.0	86.5	70.6	84.8	11.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.1	116.7	81.6	67.4	110.1	79.6	66.0	104.6	78.4	38.8	87.9	37.4	86.8	36.1	85.0	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.9	26.5	23.2	DB	27.1	100.1	3.7	2.7	24.4	102.0	22.2	103.6	20.1	105.1	17.4	107.1
			WB	23.9	76.8	2.6	2.1	22.0	78.3	20.5	79.5	19.0	80.7	17.1	82.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	77.0	74.4	70.3	63.2	57.0	52.2	51.4	55.5	60.5	66.4	71.3	75.4
	DBStd	11.12	5.43	6.22	6.12	6.58	6.54	5.95	6.95	7.90	8.07	7.49	7.03	6.30
	HDD50	178	0	0	0	4	15	45	65	34	13	2	0	0
	HDD65	1790	2	7	23	107	256	383	422	311	177	73	23	6
	CDD50	5474	837	685	630	399	233	111	110	203	328	510	640	788
	CDD65	1611	373	272	187	53	9	0	2	15	42	116	213	329
	CDH74	15792	3902	2596	1508	394	59	5	23	191	425	1093	2118	3478
	CDH80	6243	1754	1069	479	65	4	0	3	51	109	350	819	1540
Wind	WSAvg	10.2	11.1	10.2	9.7	8.6	7.9	8.1	8.7	10.6	11.4	12.8	12.2	11.6
Precipitation	PrecAvg	24.8	4.3	3.9	3.6	1.6	0.6	0.3	0.2	0.4	0.8	1.7	3.1	4.0
	PrecMax	38.8	8.4	14.5	8.0	5.8	4.2	3.0	1.1	2.2	3.7	5.0	7.4	9.0
	PrecMin	9.1	0.4	1.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0
	PrecStd	5.9	2.0	2.4	2.0	1.5	0.9	0.5	0.2	0.5	0.9	1.3	1.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.0	96.2	91.7	85.7	79.1	74.2	76.9	85.6	88.4	92.3	94.5	98.2
		MCWB	70.0	70.3	69.8	67.3	62.1	56.6	58.4	59.9	61.8	64.6	67.0	70.5
	2%	DB	94.9	92.3	88.1	81.7	75.3	69.9	71.4	79.3	82.9	87.5	91.2	94.6
		MCWB	69.9	69.6	68.1	64.6	60.7	54.7	54.1	56.5	58.9	62.8	65.4	68.8
	5%	DB	91.7	89.6	85.5	78.7	72.1	66.5	67.6	74.0	78.9	83.8	88.1	91.5
		MCWB	68.9	69.5	67.3	62.6	58.7	52.9	51.9	54.5	57.0	60.7	64.4	67.7
	10%	DB	88.9	86.9	82.5	75.4	68.8	63.5	63.9	69.5	75.1	80.3	84.9	88.2
		MCWB	68.3	68.2	66.3	61.4	57.1	51.3	49.8	52.2	55.3	59.3	63.4	66.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.7	75.4	73.6	70.8	65.1	60.9	60.7	62.8	64.2	68.7	71.3	74.3
		MCDB	90.5	88.0	86.7	81.1	74.3	69.6	72.9	80.3	83.9	85.7	88.7	90.2
	2%	WB	73.2	73.4	71.3	67.6	62.7	57.2	57.0	59.2	61.5	65.4	68.5	72.0
		MCDB	88.5	86.7	84.0	76.0	71.9	66.3	67.2	74.6	78.3	80.8	84.6	88.4
	5%	WB	71.6	71.7	69.3	65.5	60.9	54.9	54.1	56.2	58.9	63.4	66.6	70.4
		MCDB	86.7	85.0	80.8	74.2	70.0	63.4	63.7	70.4	74.8	78.7	83.7	86.4
	10%	WB	70.1	70.0	67.5	63.4	58.7	52.7	51.5	53.4	56.7	61.5	64.9	68.6
		MCDB	84.8	83.0	78.9	72.9	66.5	61.1	61.8	67.7	71.5	76.3	81.3	84.1
Mean Daily Temperature Range	5% DB	MDBR	20.8	20.2	18.8	18.4	17.9	19.3	20.5	21.3	21.0	20.6	21.3	21.4
		MCDBR	23.7	23.3	22.5	23.0	21.9	23.5	25.3	26.8	25.2	24.6	24.2	24.4
		MCWBR	9.6	9.6	10.0	10.9	11.1	12.6	13.0	13.0	11.8	11.1	10.5	10.6
	5% WB	MCDBR	20.7	20.2	19.7	18.5	18.7	20.4	22.3	24.1	22.5	21.8	22.1	21.9
		MCWBR	10.2	9.6	9.6	10.7	10.6	12.2	13.0	13.5	11.8	11.2	10.9	10.9
Clear-Sky Solar Irradiance	taub	0.372	0.366	0.352	0.343	0.315	0.302	0.303	0.332	0.376	0.365	0.363	0.370	
	taud	2.430	2.449	2.485	2.479	2.509	2.548	2.501	2.405	2.275	2.363	2.398	2.409	
	Ebn at Noon	306	302	295	279	271	268	273	278	281	298	307	308	
	Edh at Noon	39	37	33	30	25	23	25	31	40	40	40	40	
All-Sky Solar Radiation	RadAvg	2425	2110	1716	1269	931	873	956	1255	1632	1999	2350	2504	
	RadStd	125	138	144	134	107	75	62	92	135	172	132	124	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+1.03	N/A	+0.77	+1.46	+1.48	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	+1.01	+1.12	+1.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page